Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production software engineering in **housing, leasing, and affordable-housing compliance**, specializing in agentic AI systems that automate real-world property workflows. Built LeaseI — an agentic LangGraph pipeline auditing 1,240 residential lease PDFs for compliance gaps across 10 rules, 5 US states + federal standards (93.6% precision, 87.9% F1). Built AgentProbe — an open-source multi-agent AI testing toolkit (LangGraph, LLM-as-judge eval engine, autonomous agents). Shipped LIHTC/Tax Credit, Section 8/HCV, and HUD compliance modules at scale for 100K+ unit portfolios at Entrata. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Developed **affordable housing compliance automation** (LIHTC/Tax Credit, Section 8/HCV, HUD) — income eligibility verification against AMI thresholds, recertification workflows, and waitlist management across **100K+ unit portfolios**; eliminated 80% of manual operational workload via Python/PL-SQL automation; zero compliance violations.
- Built and owned **leasing and property management workflows at scale** — 20+ production modules (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records; 99.5% uptime over 6+ months.
- Pioneered Entrata's AI engineering foundation — built and shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Cut microservices execution time 50% via Docker containerization; improved query performance 30% across 200+ SQL queries; achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); led 300+ peer code reviews.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · ML Research · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%; translated complex model outputs into actionable insights for non-technical stakeholders.

PROJECTS

LeaseI: Agentic AI Lease & Affordable-Housing Compliance System

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph pipeline enabling autonomous compliance audits over 1,240 residential lease PDFs — gap detection across 10 rules, 5 US states + federal affordable-housing standards — **93.6% precision, 82.9% recall, 87.9% F1**; deterministic rules engine keeps every compliance decision fully auditable.
- Implemented pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with regression gate; LLM-as-judge scores both accuracy and reasoning quality — same eval discipline needed in production AI for housing workflows at scale.

AgentProbe: Multi-Agent AI Testing & Evaluation Toolkit

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing — autonomous agents that identify issues, trace retrieval quality, and self-correct without human intervention.
- Implemented **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD quality gates; React dashboard tracking per-run metrics, regression drift, and latency across concurrent agent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries to FHIR APIs using LLMs, BERT, and semantic embeddings — **HIPAA-compliant RBAC** across structured medical records; same agentic NLP-over-documents pattern used in LeaseI.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, JavaScript, TypeScript, R, Bash, HTML/CSS

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, LLM-as-Judge, LLM Evaluation, Prompt Engineering, ChromaDB, OpenAI API, Anthropic API, Embeddings, Semantic Search, NLP, spaCy, BERT, FastAPI, REST APIs

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, NumPy, Pandas, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), ETL, Data Pipelines, Microservices, RabbitMQ, PostgreSQL, MySQL, Git, Linux, Agile/Scrum